

Distance-Based Outlier Detection

□ Let r ($r \geq 0$) be a distance threshold and π ($0 < \pi \leq 1$) be a fraction threshold

□ An object, o , is a $DB(r, \pi)$ -outlier if

$$\frac{|\{o' | dist(o, o') \leq r\}|}{|D|} \leq \pi$$

Algorithm: Distance-based outlier detection.

Input:

- a set of objects $D = \{o_1, \dots, o_n\}$, threshold r ($r > 0$) and π ($0 < \pi \leq 1$);

Output: $DB(r, \pi)$ outliers in D .

Method:

```
for  $i = 1$  to  $n$  do
   $count \leftarrow 0$ 
  for  $j = 1$  to  $n$  do
    if  $i \neq j$  and  $dist(o_i, o_j) \leq r$  then
       $count \leftarrow count + 1$ 
      if  $count \geq \pi \cdot n$  then
        exit { $o_i$  cannot be a  $DB(r, \pi)$  outlier}
      endif
    endif
  endfor
  print  $o_i$  { $o_i$  is a  $DB(r, \pi)$  outlier according to (Eq. 11.10)}
endfor;
```